UCS748: Generative AI

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2	0	2	3

Course Objective: This course introduces students to the field of generative artificial intelligence with a focus on Large Language Models (LLMs). Students will learn the theoretical foundations behind LLMs and gain hands-on experience in training and fine-tuning these models for various generative tasks such as text generation, image generation, and more.

Syllabus

Introduction to Generative AI: Generative AI: Meaning, Capabilities and Potential, Applications of Generative AI, Tools for text, images, videos, audios and code generation.

Generative AI Models: Introduction to Large Language Models (LLMs), History and evolution of LLMs, Transformer architecture: Attention mechanism, Pre-training and fine-tuning of LLMs, Language modeling objectives (e.g., Masked Language Modeling, Next Sentence Prediction), Data preprocessing for LLMs, Training strategies and best practices, Fine-tuning on custom datasets, Handling domain-specific data and tasks.

Text Generation with LLMs: Text generation techniques, Conditional text generation, Sampling strategies, Evaluation metrics for text generation

Image Generation with LLMs: Overview of image generation tasks, Generative Adversarial Networks (GANs) vs. LLMs for image generation, Fine-tuning LLMs for image generation, Evaluation metrics for image generation

Beyond Text and Images: Multi-Modal Generation: Introduction to multi-modal generation, Combining LLMs with other generative models, Applications of multi-modal generation.

Prompt Engineering: Meaning of Prompts and Prompt Engineering, Text-to-text prompt techniques: Interview Pattern Approach, Chain-of-Thought Approach, Tree-of-Thought Approach.

Laboratory Work:

1. Pre-train a small language model on a text corpus and fine-tune it on a specific task or dataset

2. Fine-tune pre-trained language models on custom datasets for specific tasks like sentiment analysis or text classification.

3. Implement autoregressive decoding to generate text using a pre-trained language model and explore its limitations.

4. Build a conditional text generation model that takes input prompts or contexts to generate relevant responses.

5. Build a basic GAN architecture using a deep learning framework like TensorFlow or PyTorch.

6. Explore techniques to fine-tune pre-trained language models like GPT for image generation tasks using frameworks like CLIP.

7. Experiment with combining LLMs like GPT with other generative models like GANs or Variational Autoencoders (VAEs) for multi-modal generation tasks.

8. Implement a multi-modal generation model that generates coherent captions for given images or generates images from textual descriptions.

9. Explore real-world applications of multi-modal generation such as image captioning, visual question answering (VQA), and generating visual explanations from textual input.

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Understand the theoretical foundations of Large Language Models (LLMs).

2. Apply LLMs for various generative tasks including text generation, image generation, and more.

3. Train and fine-tune LLMs on custom datasets.

4. Explore advanced topics and applications of LLMs in research and industry.

Text Books

1. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
2. "Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play" by David Foster.
3. "Dive into Deep Learning" by Aston Zhang.

Reference Books:

1. "Attention is All You Need" by Ashish Vaswani et al.

2. "Natural Language Processing with PyTorch" by Delip Rao and Brian McMahan

3. "GPT-3: Language Models are Few-Shot Learners" by Brown et al.